Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Art Silva & Son Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12636 W Main ST	Crows Landing	Stanislaus	95313

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X006-X002-XXXX

B. OPERATORS

Silva, Art

Operator name: Silva, Art

Telephone no.: (209) 667-5189 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12636 West Main ST	Crows Landing	CA	95313

This operator is responsible for paying permit fees.

C. OWNERS

Silva, Art

Legal owner name: Silva, Art

Telephone no.: (209) 667-5189 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12636 West Main ST	Crows Landing	CA	95313

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	125	100	0	0
Number under roof	600	75	0	0	150	150
Maximum number	600	90	125	100	150	150
Average number	600	75	125	100	150	150
Avg live weight (lbs)	1,400	1,450	900	600		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,706.53 *tons per reporting period*

Total nitrogen from manure: 251,598.06 *lbs per reporting period*

After ammonia losses (30% loss applied): 176,118.64 *lbs per reporting period*

Total phosphorus from manure: 40,571.00 *lbs per reporting period*

Total potassium from manure: 114,905.54 *lbs per reporting period*

Total salt from manure: 299,756.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,920,000 *gallons*

Total nitrogen generated: 40,525.46 *lbs*

Total phosphorus generated: 5,900.21 *lbs*

Total potassium generated: 52,562.28 *lbs*

Total salt generated: 323,916.85 *lbs*

	7,920,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,920,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 4	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/21/2023	Solid commercial fertilizer *NH3*	5.80 *ton*	Dry-weight	0.1	82.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	9,502.49	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	9,502.49	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	Corral solids	960.00 *ton*	Dry-weight	12.8		21,700.00	7,250.00	24,300.00		50.80
10/01/2023	Corral solids	1,560.00 *ton*	Dry-weight	12.8		21,700.00	7,250.00	24,300.00		50.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	95,368.90	31,862.88	106,795.58	2,232,599.04
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	95,368.90	31,862.88	106,795.58	2,232,599.04

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Barrera	14	6	0	none	X058-X003-X003-XXXX
Field 1	40	40	2	process wastewater	X058-X006-X002-XXXX
Field 2	15	15	2	process wastewater	X058-X006-X002-XXXX
Field 3	8	4	2	process wastewater	X058-X006-X002-XXXX
Field 4	19	19	1	process wastewater	X058-X004-X017-XXXX
Field 5	19	19	1	process wastewater	X058-X004-X016-XXXX
Field 6	19	19	2	process wastewater	X058-X004-X003-XXXX
Field 7	45	45	2	manure	X058-X003-X031-XXXX
Field 8	55	55	2	manure	X058-X001-X003-XXXX
Field 9	55	55	2	manure	X058-X001-X013-XXXX
Totals for areas that were used for application	275	271	16		
Totals for areas that were not used for application	14	6	0		
Land application area totals	289	277	16		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	620.00 *ton*	As-is		70.0	4,900.00	1,120.00	7,140.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.50	151.90	34.72	221.34	930.00

Field 1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	925.00 *ton*	As-is		70.0	3,820.00	930.00	6,000.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	23.13	176.68	43.01	277.50	1,803.75

Field 2

Field name: Field 2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	270.00 *ton*	As-is		70.0	4,440.00	1,050.00	5,410.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	18.00	159.84	37.80	194.76	1,090.80

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	365.00 *ton*	As-is		70.0	3,670.00	900.00	5,700.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.33	178.61	43.80	277.40	2,058.60

Field 3

Field name: Field 3

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	70.00 *ton*	As-is		70.0	4,340.00	1,100.00	7,490.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	17.50	151.90	38.50	262.15	1,060.50

05/15/2023: Sorghum-Sudangrass, forage

Crop: Sorghum-Sudangrass, forage Acres planted: 4 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	50.00 *ton*	As-is		70.0	7,980.00	1,140.00	13,800.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	656.00	112.00	784.00	0.00
Total actual harvest content	12.50	199.50	28.50	345.00	960.00

Field 4

Field name: Field 4

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	300.00 *ton*	As-is		70.0	4,330.00	1,050.00	5,770.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.79	136.74	33.16	182.21	956.84

Field 5

Field name: Field 5

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	300.00 *ton*	As-is		70.0	5,100.00	1,150.00	6,870.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.79	161.05	36.32	216.95	937.89

Field 6

Field name: Field 6

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	300.00 *ton*	As-is		70.0	5,170.00	1,110.00	7,200.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.79	163.26	35.05	227.37	1,089.47

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	425.00 *ton*	As-is		70.0	3,970.00	840.00	6,000.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	22.37	177.61	37.58	268.42	671.05

Field 7

Field name: Field 7

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	750.00 *ton*	As-is		70.0	4,440.00	910.00	7,490.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.67	148.00	30.33	249.67	1,370.00

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,100.00 *ton*	As-is		70.0	3,700.00	960.00	6,600.00		6.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.44	180.89	46.93	322.67	951.87

Field 8

Field name: Field 8

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	900.00 *ton*	As-is		70.0	4,240.00	970.00	5,650.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.36	138.76	31.75	184.91	1,305.82

Field 8

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,275.00 *ton*	As-is		70.0	3,590.00	870.00	6,000.00		6.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.18	166.45	40.34	278.18	958.34

Field 9

Field name: Field 9

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	900.00 *ton*	As-is		70.0	4,270.00	950.00	5,910.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.36	139.75	31.09	193.42	1,207.64

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,275.00 *ton*	As-is		70.0	3,610.00	930.00	6,300.00		6.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.18	167.37	43.12	292.09	849.85

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/03/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.71	4.76	76.01	400.56	480,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,050,000.00 *gal*
Application event totals		80.71	4.76	76.01	422.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.42	9.93	85.72	421.39	384,000.00 *gal*
Application event totals		72.42	9.93	85.72	421.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.32	7.45	64.29	316.04	288,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.18	2,430,000.00 *gal*
Application event totals		54.32	7.45	64.29	329.22	

Field 1 - 06/01/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	127.08	16.52	181.75	1,090.52	720,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	32.95	6,075,000.00 *gal*
Application event totals		127.08	16.52	181.75	1,123.48	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.36	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.59	9.25	58.16	458.24	528,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	24.17	4,455,000.00 *gal*
Application event totals		34.59	9.25	58.16	482.41	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.44	8.41	52.87	416.58	480,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,050,000.00 *gal*
Application event totals		31.44	8.41	52.87	438.55	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Field 1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.30	7.57	47.59	374.92	432,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.77	3,645,000.00 *gal*
Application event totals		28.30	7.57	47.59	394.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.30	7.57	47.59	374.92	432,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.77	3,645,000.00 *gal*
Application event totals		28.30	7.57	47.59	394.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Field 2 - 11/04/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.62	6.34	101.34	534.08	240,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,025,000.00 *gal*
Application event totals		107.62	6.34	101.34	563.37	

Field 2 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.56	13.25	114.29	561.85	192,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		96.56	13.25	114.29	585.28	

Field 2 - 06/01/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.96	14.69	161.56	969.36	240,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	35.15	2,430,000.00 *gal*
Application event totals		112.96	14.69	161.56	1,004.50	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.54	8.97	56.40	444.35	192,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		33.54	8.97	56.40	467.79	

Field 2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.54	8.97	56.40	444.35	192,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		33.54	8.97	56.40	467.79	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.54	8.97	56.40	444.35	192,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		33.54	8.97	56.40	467.79	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.54	8.97	56.40	444.35	192,000.00 *gal*
Application event totals		33.54	8.97	56.40	444.35	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	26.36	

Field 3 - 11/04/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.46	18.63	192.27	1,030.44	72,000.00 *gal*
Application event totals		114.46	18.63	192.27	1,030.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.12	8.45	64.49	598.84	48,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		89.12	8.45	64.49	620.81	

Field 3 - 05/15/2023: Sorghum-Sudangrass, forage

Field name: Field 3

Crop: Sorghum-Sudangrass, forage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	32.95	607,500.00 *gal*
Application event totals		0.00	0.00	0.00	32.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Field 3 - 05/15/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	127.08	16.52	181.75	1,090.52	72,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	32.95	607,500.00 *gal*
Application event totals		127.08	16.52	181.75	1,123.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.44	8.41	52.87	416.58	48,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		31.44	8.41	52.87	438.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.44	8.41	52.87	416.58	48,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		31.44	8.41	52.87	438.55	

Field 3 - 05/15/2023: Sorghum-Sudangrass, forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.44	8.41	52.87	416.58	48,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		31.44	8.41	52.87	438.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.44	8.41	52.87	416.58	48,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	405,000.00 *gal*
Application event totals		31.44	8.41	52.87	438.55	

Field 4 - 11/05/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.96	5.01	80.01	421.64	240,000.00 *gal*
Application event totals		84.96	5.01	80.01	421.64	

Field 4 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.23	10.46	90.23	443.57	192,000.00 *gal*
Application event totals		76.23	10.46	90.23	443.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.75	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.75	

Field 5 - 11/05/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.96	5.01	80.01	421.64	240,000.00 *gal*
Application event totals		84.96	5.01	80.01	421.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.23	10.46	90.23	443.57	192,000.00 *gal*
Application event totals		76.23	10.46	90.23	443.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	10.67	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.67	

Field 6 - 11/04/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.23	10.46	90.23	443.57	192,000.00 *gal*
Application event totals		76.23	10.46	90.23	443.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.75	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.95	6.01	96.01	505.97	288,000.00 *gal*
Application event totals		101.95	6.01	96.01	505.97	

Field 6 - 06/15/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.85	16.23	178.57	1,071.39	336,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	32.37	2,835,000.00 *gal*
Application event totals		124.85	16.23	178.57	1,103.77	

Field 6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.75	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.75	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.12	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.10	8.85	55.66	438.51	240,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		33.10	8.85	55.66	461.63	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.10	8.85	55.66	438.51	240,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		33.10	8.85	55.66	461.63	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.12	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.12	

Field 6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.48	7.08	44.53	350.81	192,000.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	1,620,000.00 *gal*
Application event totals		26.48	7.08	44.53	369.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.50	

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	165.26	54.42	87.17	1,567.81	600.00 *ton*
Application event totals		165.26	54.42	87.17	1,567.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

Field 7 - 06/15/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	173.48	66.17	1.79	5,114.95	400.00 *ton*
Application event totals		173.48	66.17	1.79	5,114.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	35.15	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	31.24	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	61.50	0.00	0.00	0.00	
Application event totals		61.50	0.00	0.00	0.00	

Field 7 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.34	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.39	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.34	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.43	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.43	

Field 8 - 11/02/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 8 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	169.01	55.66	89.15	1,603.45	750.00 *ton*
Application event totals		169.01	55.66	89.15	1,603.45	

Field 8 - 06/16/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	177.42	67.68	1.83	5,231.20	500.00 *ton*
Application event totals		177.42	67.68	1.83	5,231.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	38.34	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.34	

Field 8 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	31.95	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	30.36	7,695,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	61.50	0.00	0.00	0.00	
Application event totals		61.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.16	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.56	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	22.37	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.37	

Field 8 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.56	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

Field 9 - 11/02/2022: Oats, silage-soft dough

Field name: Field 9

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Separator solids	169.01	55.66	89.15	1,603.45	750.00 *ton*
Application event totals		169.01	55.66	89.15	1,603.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

Field 9 - 06/17/2023: Corn, silage

Field name: Field 9

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	177.42	67.68	1.83	5,231.20	500.00 *ton*
Application event totals		177.42	67.68	1.83	5,231.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	38.34	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	31.95	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	28.76	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	61.50	0.00	0.00	0.00	
Application event totals		61.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	28.76	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.76	

Field 9 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	28.76	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	27.16	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.56	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.56	

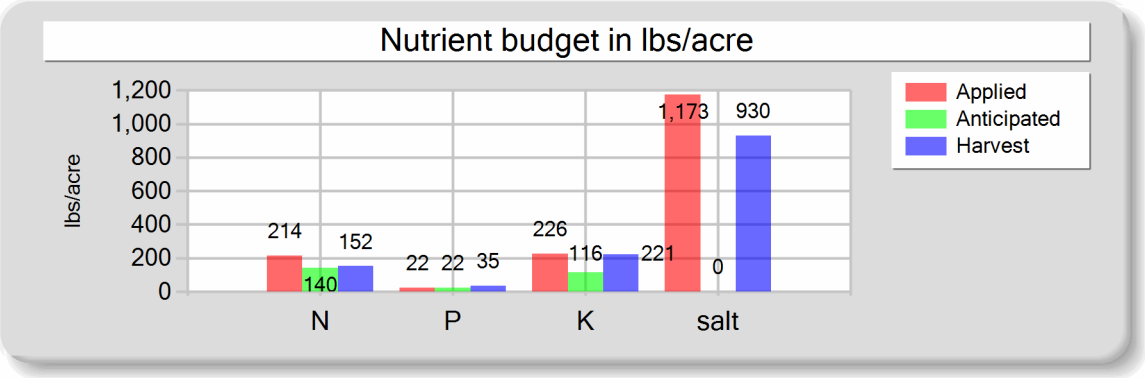
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	23.97	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.97	

B. NUTRIENT BUDGET

Field 1 - 11/03/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.45	22.14	226.02	1,137.99
Fresh water	0.00	0.00	0.00	35.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.45	22.14	226.02	1,173.14
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	151.90	34.72	221.34	930.00
Nutrient balance	62.55	-12.58	4.68	243.14
Applied to removed ratio	1.41	0.64	1.02	1.26

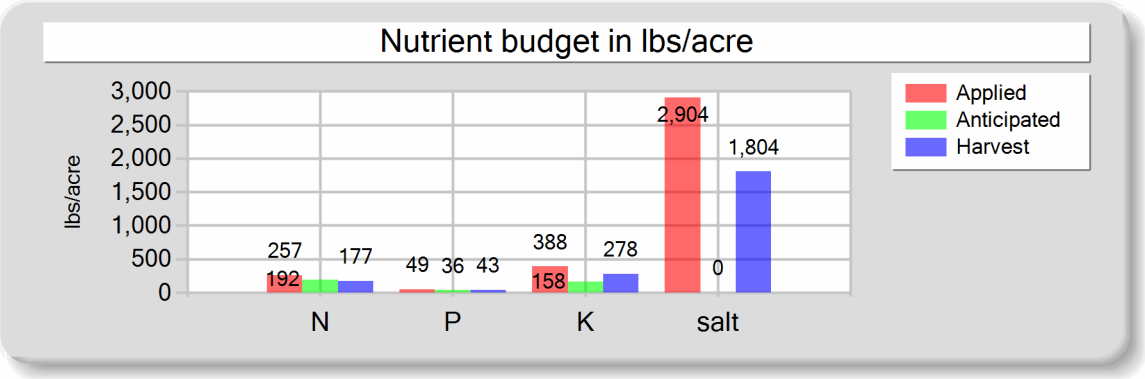
Fresh water applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
5.97	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/01/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	249.71	49.33	387.96	2,715.20
Fresh water	0.00	0.00	0.00	188.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.71	49.33	387.96	2,904.12
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	176.68	43.01	277.50	1,803.75
Nutrient balance	80.03	6.32	110.46	1,100.37
Applied to removed ratio	1.45	1.15	1.40	1.61

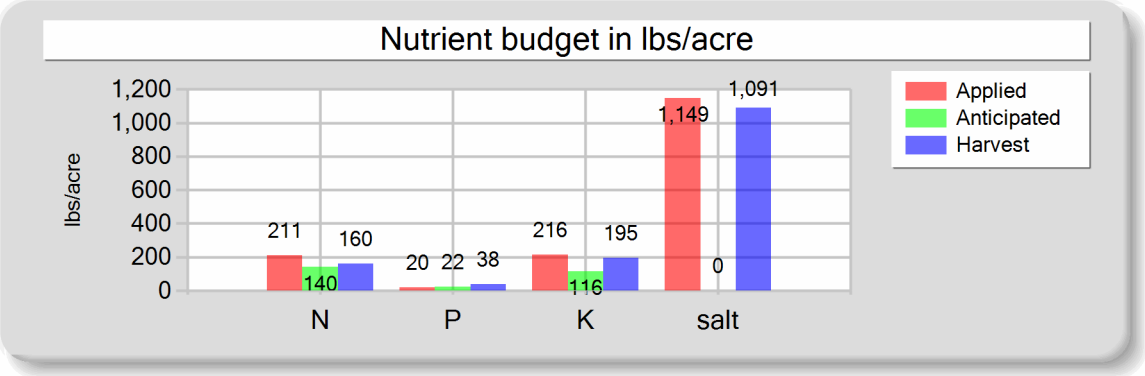
Fresh water applied	
34,830,000.00	*gallons*
1,282.67	*acre-inches*
32.07	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
2.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/04/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.18	19.59	215.63	1,095.93
Fresh water	0.00	0.00	0.00	52.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.18	19.59	215.63	1,148.66
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	159.84	37.80	194.76	1,090.80
Nutrient balance	51.34	-18.21	20.87	57.86
Applied to removed ratio	1.32	0.52	1.11	1.05

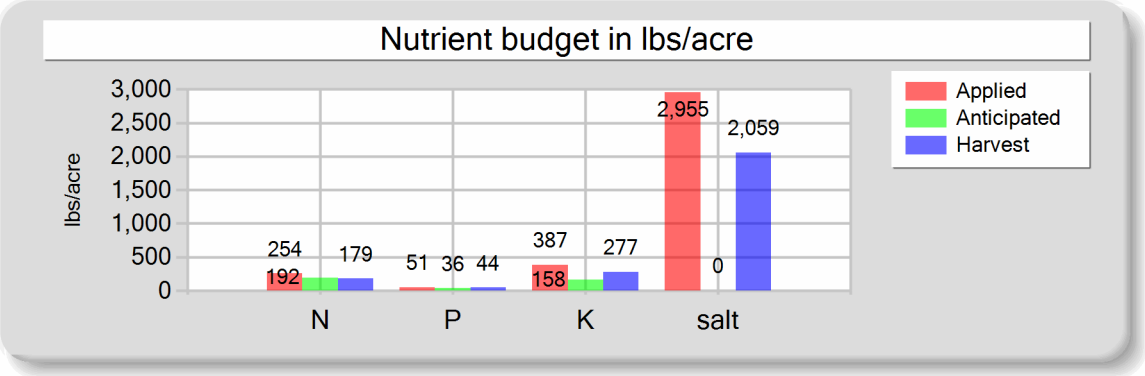
Fresh water applied	
3,645,000.00	*gallons*
134.23	*acre-inches*
8.95	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/01/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	247.12	50.58	387.15	2,746.77
Fresh water	0.00	0.00	0.00	207.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.12	50.58	387.15	2,954.74
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	178.61	43.80	277.40	2,058.60
Nutrient balance	75.51	6.78	109.75	896.14
Applied to removed ratio	1.42	1.15	1.40	1.44

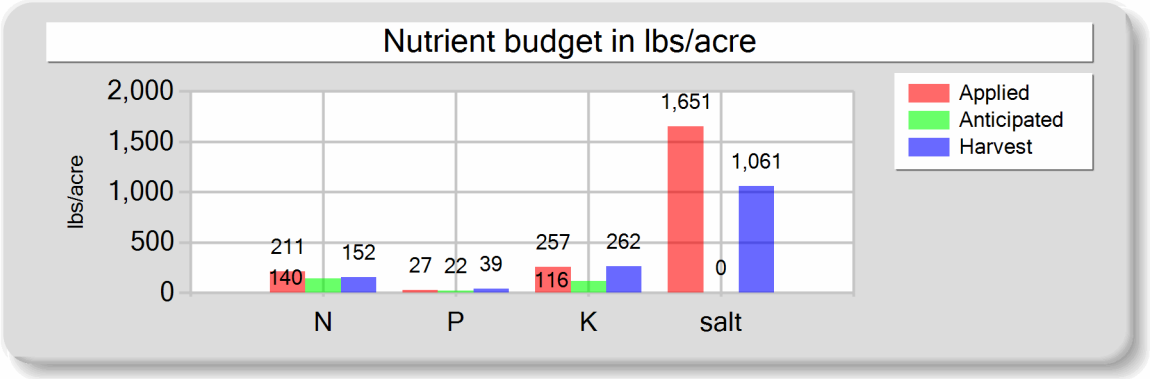
Fresh water applied	
14,377,500.00	*gallons*
529.47	*acre-inches*
35.30	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
2.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/04/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.58	27.08	256.76	1,629.28
Fresh water	0.00	0.00	0.00	21.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.58	27.08	256.76	1,651.25
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	151.90	38.50	262.15	1,060.50
Nutrient balance	58.68	-11.42	-5.39	590.75
Applied to removed ratio	1.39	0.70	0.98	1.56

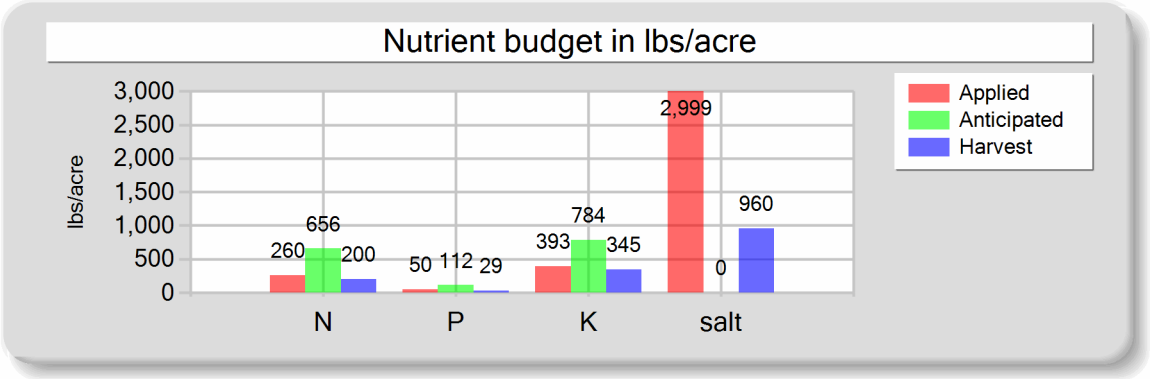
Fresh water applied	
405,000.00	*gallons*
14.91	*acre-inches*
3.73	*inches/acre*

Process wastewater applied	
120,000.00	*gallons*
4.42	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/15/2023: Sorghum-Sudangrass, forage

Field name: Field 3 Crop: Sorghum-Sudangrass, forage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.85	50.17	393.25	2,756.85
Fresh water	0.00	0.00	0.00	241.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.85	50.17	393.25	2,998.50
Anticipated crop nutrient removal	656.00	112.00	784.00	0.00
Actual crop nutrient removal	199.50	28.50	345.00	960.00
Nutrient balance	60.35	21.67	48.25	2,038.50
Applied to removed ratio	1.30	1.76	1.14	3.12

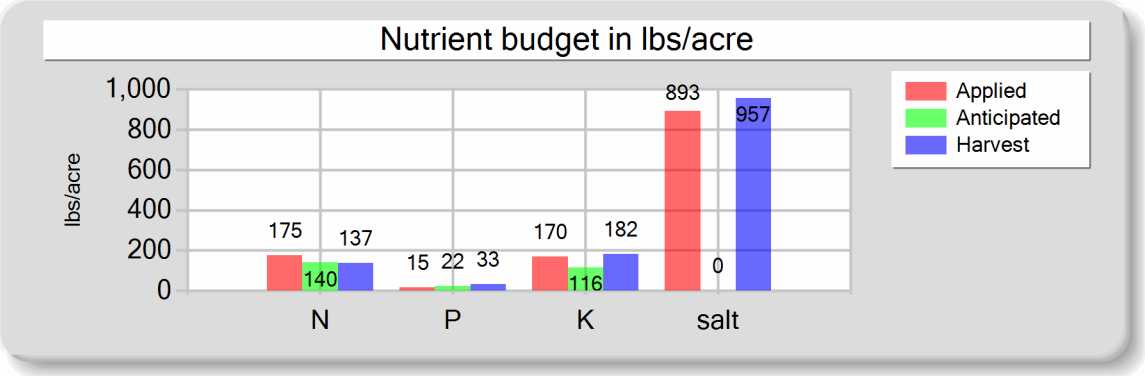
Fresh water applied	
4,455,000.00	*gallons*
164.06	*acre-inches*
41.02	*inches/acre*

Process wastewater applied	
264,000.00	*gallons*
9.72	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/05/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.19	15.46	170.24	865.21
Fresh water	0.00	0.00	0.00	27.75
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	175.19	15.46	170.24	892.96
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	136.74	33.16	182.21	956.84
Nutrient balance	38.46	-17.69	-11.97	-63.88
Applied to removed ratio	1.28	0.47	0.93	0.93

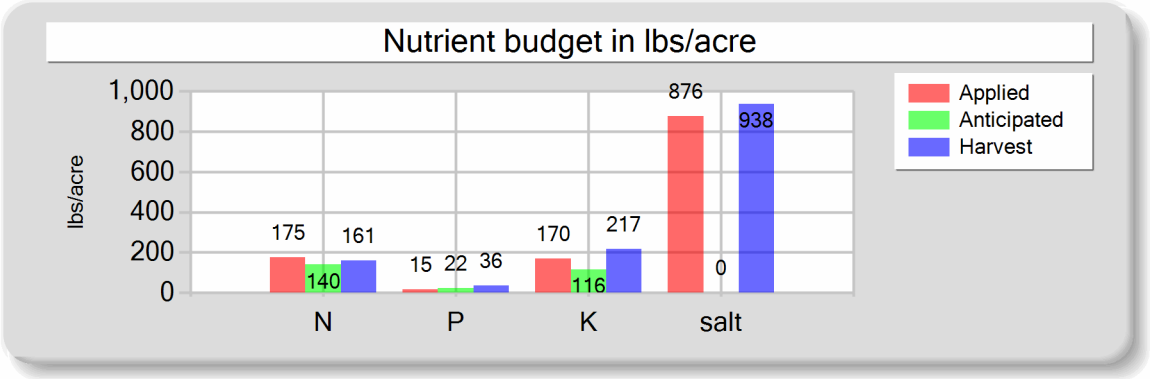
Fresh water applied	
2,430,000.00	*gallons*
89.49	*acre-inches*
4.71	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/05/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.19	15.46	170.24	865.21
Fresh water	0.00	0.00	0.00	10.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	175.19	15.46	170.24	875.88
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	161.05	36.32	216.95	937.89
Nutrient balance	14.14	-20.85	-46.71	-62.01
Applied to removed ratio	1.09	0.43	0.78	0.93

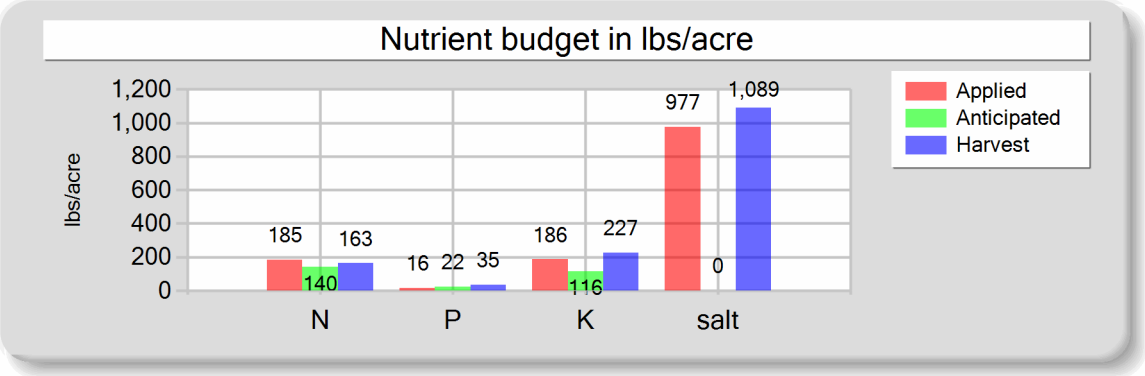
Fresh water applied	
2,430,000.00	*gallons*
89.49	*acre-inches*
4.71	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/04/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.19	16.47	186.24	949.54
Fresh water	0.00	0.00	0.00	27.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.19	16.47	186.24	977.29
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	163.26	35.05	227.37	1,089.47
Nutrient balance	21.92	-18.59	-41.13	-112.19
Applied to removed ratio	1.13	0.47	0.82	0.90

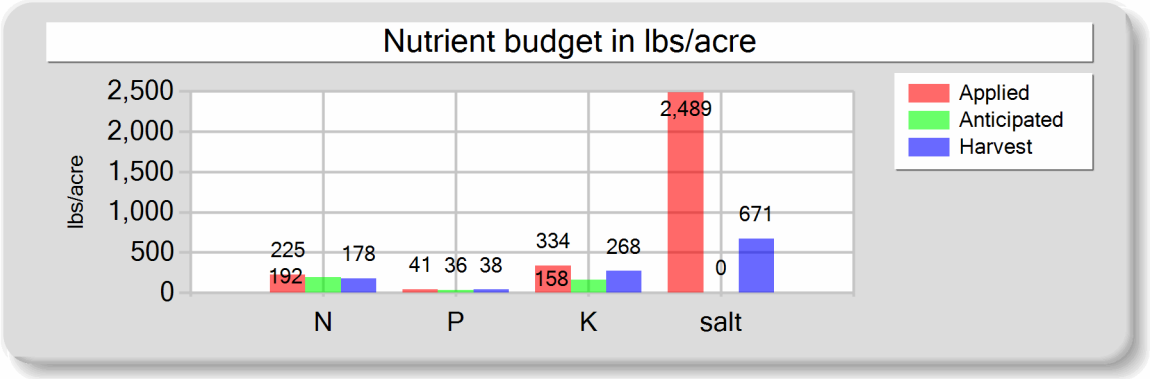
Fresh water applied	
2,430,000.00	*gallons*
89.49	*acre-inches*
4.71	*inches/acre*

Process wastewater applied	
480,000.00	*gallons*
17.68	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 06/15/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.53	41.03	334.40	2,299.21
Fresh water	0.00	0.00	0.00	189.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.53	41.03	334.40	2,488.83
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	177.61	37.58	268.42	671.05
Nutrient balance	46.92	3.45	65.98	1,817.78
Applied to removed ratio	1.26	1.09	1.25	3.71

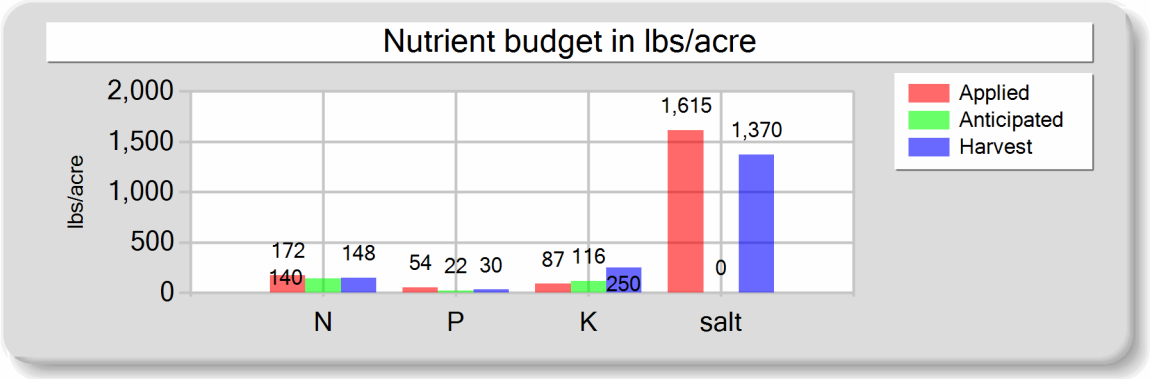
Fresh water applied	
16,605,000.00	*gallons*
611.51	*acre-inches*
32.18	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/01/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	165.26	54.42	87.17	1,567.81
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	46.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.26	54.42	87.17	1,614.68
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	148.00	30.33	249.67	1,370.00
Nutrient balance	24.26	24.09	-162.50	244.68
Applied to removed ratio	1.16	1.79	0.35	1.18

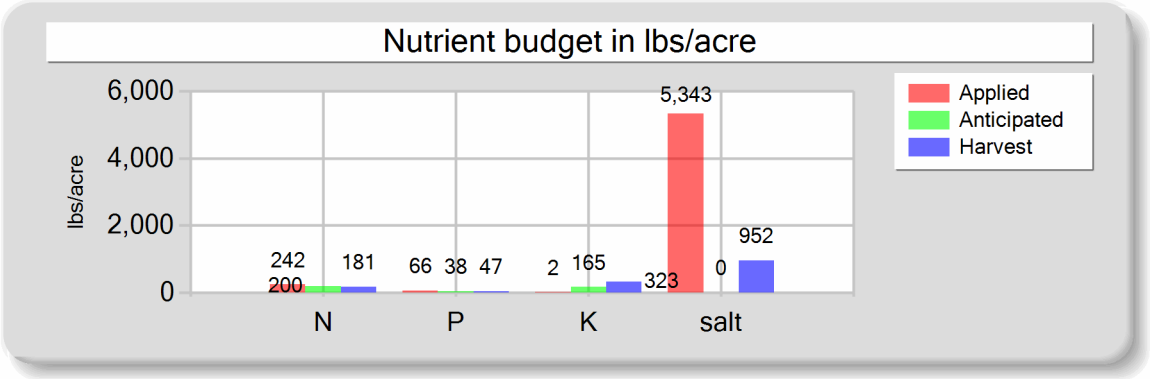
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
7.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 06/15/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	61.50	0.00	0.00	0.00
Dry manure	173.48	66.17	1.79	5,114.95
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	228.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.98	66.17	1.79	5,343.42
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	180.89	46.93	322.67	951.87
Nutrient balance	61.09	19.24	-320.88	4,391.55
Applied to removed ratio	1.34	1.41	0.01	5.61

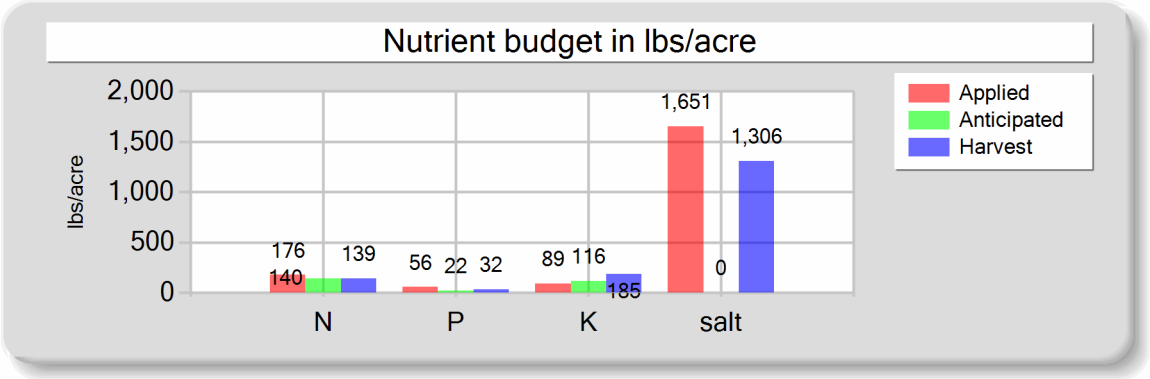
Fresh water applied	
47,385,000.00	*gallons*
1,745.03	*acre-inches*
38.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/02/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.01	55.66	89.15	1,603.45
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	47.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.01	55.66	89.15	1,651.38
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	138.76	31.75	184.91	1,305.82
Nutrient balance	37.25	23.91	-95.76	345.56
Applied to removed ratio	1.27	1.75	0.48	1.26

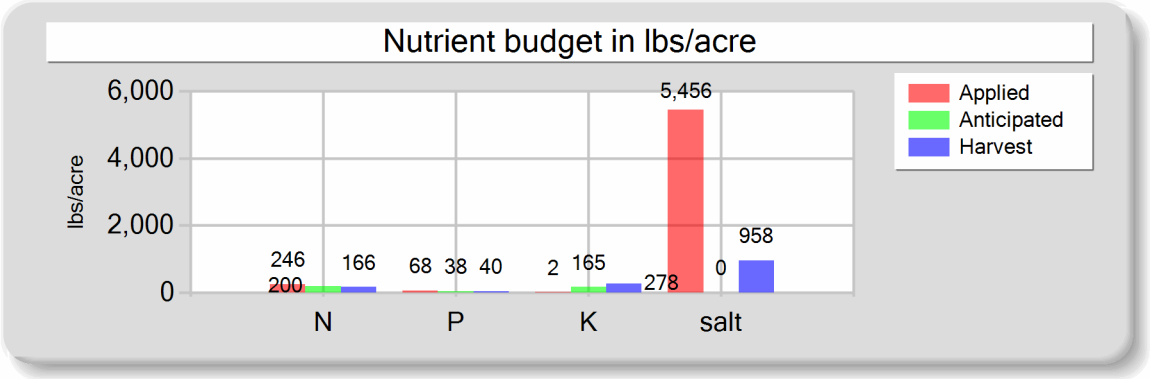
Fresh water applied	
12,150,000.00	*gallons*
447.44	*acre-inches*
8.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 06/16/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	61.50	0.00	0.00	0.00
Dry manure	177.42	67.68	1.83	5,231.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	225.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.92	67.68	1.83	5,456.47
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	166.45	40.34	278.18	958.34
Nutrient balance	79.48	27.34	-276.35	4,498.14
Applied to removed ratio	1.48	1.68	0.01	5.69

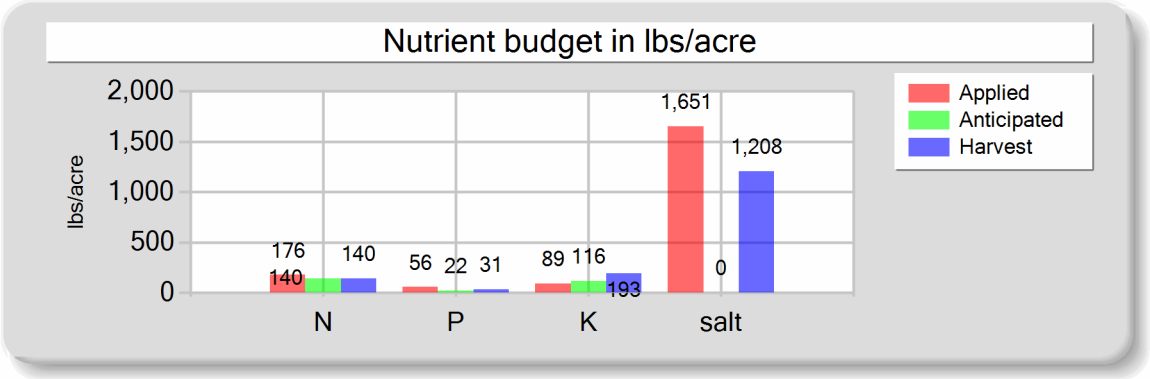
Fresh water applied	
57,105,000.00	*gallons*
2,102.98	*acre-inches*
38.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 11/02/2022: Oats, silage-soft dough

Field name: Field 9 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.01	55.66	89.15	1,603.45
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	47.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.01	55.66	89.15	1,651.38
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	139.75	31.09	193.42	1,207.64
Nutrient balance	36.27	24.57	-104.27	443.74
Applied to removed ratio	1.26	1.79	0.46	1.37

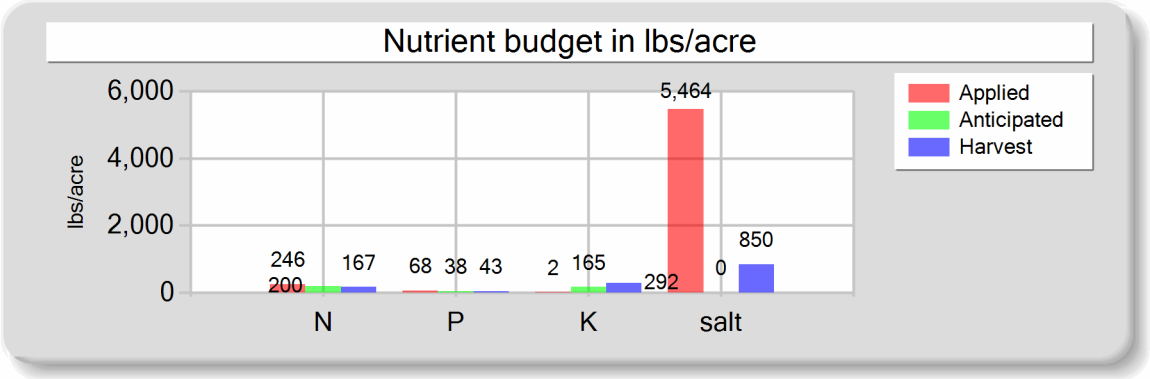
Fresh water applied	
12,150,000.00	*gallons*
447.44	*acre-inches*
8.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 06/17/2023: Corn, silage

Field name: Field 9 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	61.50	0.00	0.00	0.00
Dry manure	177.42	67.68	1.83	5,231.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	233.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.92	67.68	1.83	5,464.46
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	167.37	43.12	292.09	849.85
Nutrient balance	78.55	24.56	-290.26	4,614.62
Applied to removed ratio	1.47	1.57	0.01	6.43

Fresh water applied	
59,130,000.00	*gallons*
2,177.56	*acre-inches*
39.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,700.00	9,670.00	35,100.00							36.90
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	8,990.00	14,400.00							25.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 49.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	7,400.00	200.00							57.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,090.00	12,500.00							68.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	7,250.00	24,300.00	15,500.00	9,580.00	3,760.00	4,870.00	1,100.00		50.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	806.00	529.00	0.00	0.00	47.50	759.00								9,780.00	4,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	904.00	566.00	0.00	0.00	124.00	1,070.00								8,990.00	5,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	846.00	465.00	0.00	0.00	110.00	1,210.00								9,700.00	7,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	314.00	196.00	0.00	0.00	84.00	528.00								4,840.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 4

TID Lateral 4

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.00	26
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,120.00	7,140.00		10.00
DL	100.00	100.00	100.00		0.01

Field 1 - 06/01/2023: Corn, silage

Field 1 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,820.00	930.00	6,000.00		13.00
DL	100.00	100.00	100.00		0.01

Field 2 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,440.00	1,050.00	5,410.00		10.10
DL	100.00	100.00	100.00		0.01

Field 2 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	900.00	5,700.00		14.10
DL	100.00	100.00	100.00		0.01

Field 3 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,340.00	1,100.00	7,490.00		10.10
DL	100.00	100.00	100.00		0.01

Field 4 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,330.00	1,050.00	5,770.00		10.10
DL	100.00	100.00	100.00		0.01

Field 5 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,150.00	6,870.00		9.90
DL	100.00	100.00	100.00		0.01

Field 6 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	840.00	6,000.00		5.00
DL	100.00	100.00	100.00		0.01

Field 7 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,440.00	910.00	7,490.00		13.70
DL	100.00	100.00	100.00		0.01

Field 7 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	960.00	6,600.00		6.49
DL	100.00	100.00	100.00		0.01

Field 8 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,240.00	970.00	5,650.00		13.30
DL	100.00	100.00	100.00		0.01

Field 8 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,590.00	870.00	6,000.00		6.89
DL	100.00	100.00	100.00		0.01

Field 9 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,270.00	950.00	5,910.00		12.30
DL	100.00	100.00	100.00		0.01

Field 9 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,610.00	930.00	6,300.00		6.11
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

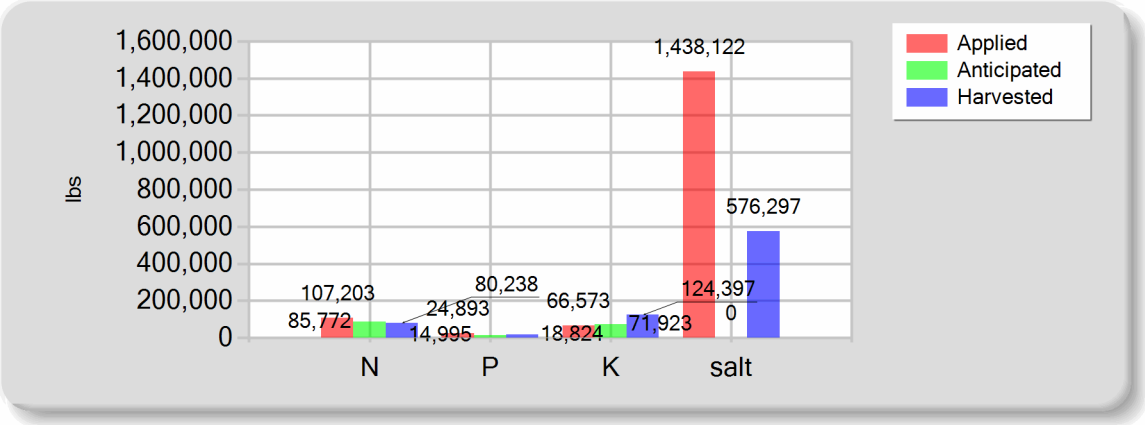
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

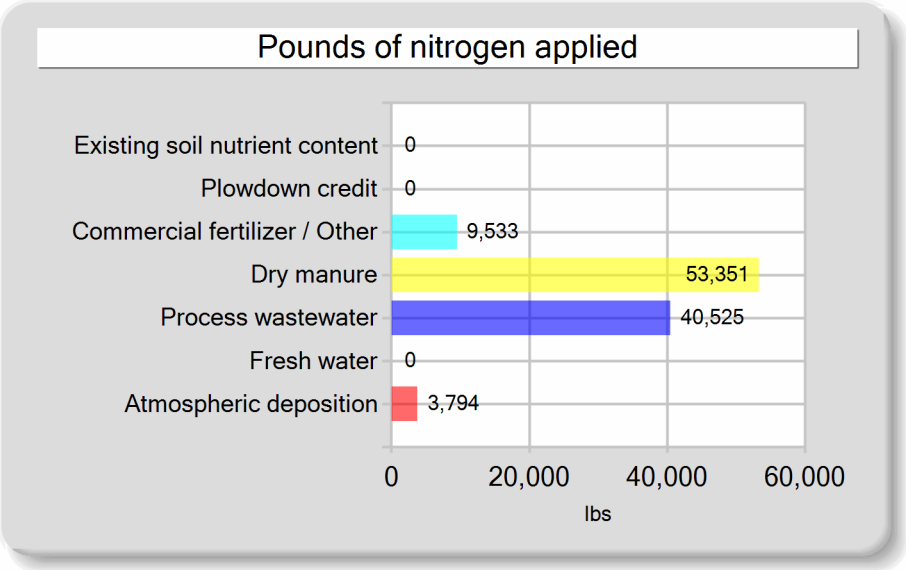
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

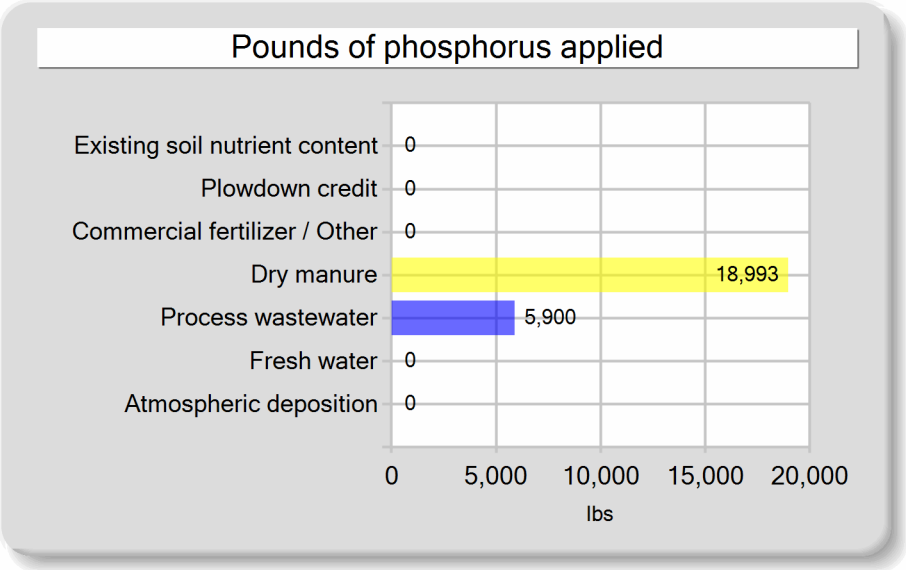
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	9,532.50	0.00	0.00	0.00
Dry manure	53,350.78	18,993.23	14,010.64	1,052,535.40
Process wastewater	40,525.46	5,900.21	52,562.28	323,916.85
Fresh water	0.00	0.00	0.00	61,669.84
Atmospheric deposition	3,794.00	0.00	0.00	0.00
Total nutrients applied	107,202.74	24,893.43	66,572.92	1,438,122.09
Anticipated crop nutrient removal	85,772.00	14,994.90	71,922.80	0.00
Actual crop nutrient removal	80,237.80	18,824.30	124,396.60	576,297.00
Nutrient balance	26,964.94	6,069.13	-57,823.68	861,825.09
Applied to removed ratio	1.34	1.32	0.54	2.50

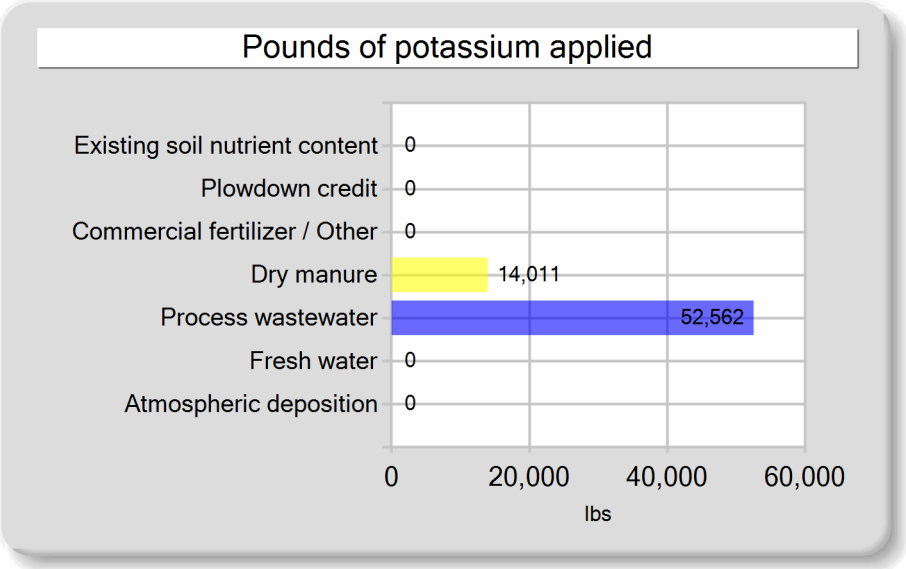
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

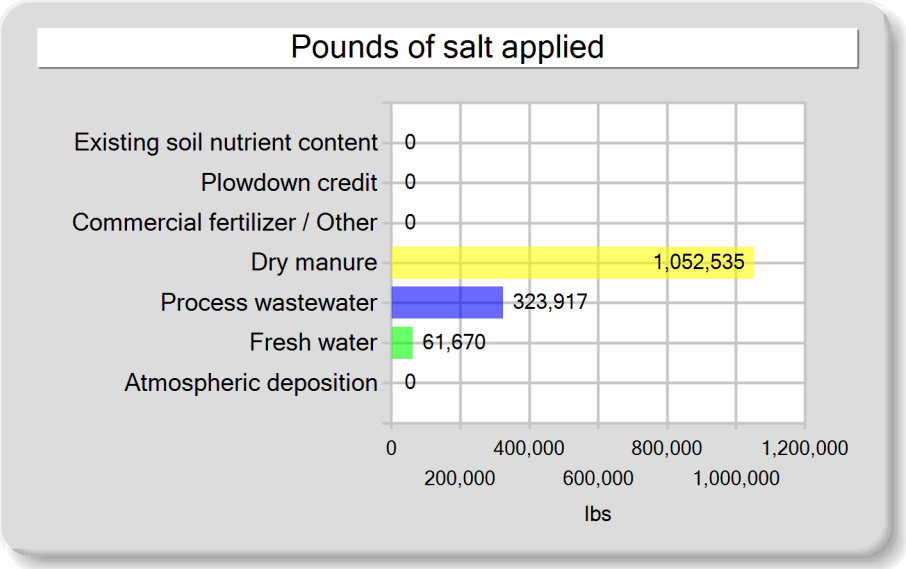


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tina Marie Silva	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Art Silva	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
4/19/2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Art Silva

Name of Dairy Facility: Art Silva & Son Dairy

Facility Address:

Number and Street	City	County	Zip Code
12636 W Main ST	Crows Landing	Stanislaus	95313

Contact Person Name and Phone Number:

Name	Phone Number
Silva	(209) 667-5189

MANURE HAULER INFORMATION

Name of Hauling Company/Person: ALP Custom Spreading Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
2324 Villa Manucha Rd	Newman	CA	95360

Contact Person:

Name	Phone Number
ALP	(209) 862-3393

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Robert Gioletti & Sons	(209) 667-6024

Address	City	State	Zip Code
10213 W. Main St	Turlock	CA	95380

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
10213 W. Main St	Turlock	95380

Street and nearest cross street (if no address)	County
	Stanislaus

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 10/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 960.00 tons

Manure Solids Content: 87.2 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Tina Marie Silva — 10/1/2023

Operator Signature — Date

Jeff Lima — 10/1/2023

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Art Silva

Name of Dairy Facility: Art Silva & Son Dairy

Facility Address:

Number and Street	City	County	Zip Code
12636 W Main ST	Crows Landing	Stanislaus	95313

Contact Person Name and Phone Number:

Name	Phone Number
Silva	(209) 667-5189

MANURE HAULER INFORMATION

Name of Hauling Company/Person: ALP Custom Spreading Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
2324 Villa Manucha Rd	Newman	CA	95360

Contact Person:

Name	Phone Number
ALP	(209) 862-3393

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Robert Gioletti & Sons	(209) 667-6024

Address	City	State	Zip Code
10213 W. Main St	Turlock	CA	95380

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
10213 W Main St	Turlock	95380

Street and nearest cross street (if no address)	County
	Stanislaus

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 10/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,560.00 tons

Manure Solids Content: 87.2 %

Method used to determine amount of manure:

Hauler Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Tina Marie Silva — 10/1/2023

Operator Signature — Date

Jeff Lima — 10/1/2023

Hauler Signature — Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Art Silva & Son Dairy
12636 W. Main St
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Barn	T3346025A	Electrical Conductivity (EC)	EPA 120.1 and	1.27	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.35	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	813	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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www.denelelabs.com

Compliance Analysis Report

Customer
Art Silva & Son Dairy
12636 W. Main St
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Dean	T3346025B	Electrical Conductivity (EC)	EPA 120.1 and	1.25	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.10	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	800	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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Compliance Analysis Report

Customer
Art Silva & Son Dairy
12636 W. Main St
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW F8	T3346025C	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	819	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Art Silva & Son Dairy
12636 W. Main St
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Buffalo	T3346025D	Electrical Conductivity (EC)	EPA 120.1 and	1.42	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	44.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	906	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Art Silva & Son Dairy
12636 W. Main St
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Mexican	T3346025E	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	820	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Art Silva & Sons
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services
Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/12	8:10	Domestic Wells	DW Barn	17.0°C		X				
2	12/12	8:15	Domestic Wells	DW Dean	17.0°C		X				
3	12/12	8:17	Domestic Wells	DW F8	15.0°C		X				
4	12/12	9:18	Domestic Wells	DW Buffalo	15.6°C		X				
5	12/12	9:23	Domestic Wells	DW Mexican	15.8°C		X				
6	12/12	8:29	wastewater	WWS@4	16.6°C			X			
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	KK	F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	12-12-23 12:00			T3346025

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)